



**AMERICAN
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**MAROUN SEMAAN FACULTY OF
ENGINEERING & ARCHITECTURE**

EECE 698/MECH 650: Autonomous Mobile Robotics

Assignment 3



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Chapter 4 — Problem 3

Effect of using two particles ($M = 2$). With two particles, the posterior is approximated as:

$$\hat{p}(x_t) = w^{(1)}\delta(x_t - x_t^{(1)}) + w^{(2)}\delta(x_t - x_t^{(2)}).$$

After resampling, the weights typically become equal, so the distribution consists of two equally weighted hypotheses. However:

- The posterior can only be represented with resolution 0.5.
- Small-probability regions may be completely ignored.
- There is a high probability of *particle depletion*, where one particle is selected twice and the other is lost.

For example, if after weighting:

$$w^{(1)} = 0.8, \quad w^{(2)} = 0.2,$$

then the probability that the higher-weight particle is selected twice during resampling is:

$$0.8^2 = 0.64,$$

which means that in 64% of the cases, the filter collapses to a single unique particle.

Bias in the posterior approximation. Yes, with $M = 2$ the posterior can be biased. Consider a discrete state space $\{A, B, C\}$ with true posterior:

$$p(x = A) = 0.6, \quad p(x = B) = 0.3, \quad p(x = C) = 0.1.$$

With two particles, the approximation can only assign probabilities in multiples of 0.5. A possible outcome after resampling is:

$$\hat{p}(x = A) = 0.5, \quad \hat{p}(x = B) = 0.5, \quad \hat{p}(x = C) = 0.$$

The bias at each state is:

$$\text{bias}(A) = 0.5 - 0.6 = -0.1,$$

$$\text{bias}(B) = 0.5 - 0.3 = +0.2,$$

$$\text{bias}(C) = 0 - 0.1 = -0.1.$$

Conclusion. Using $M = 2$ particles provides only a very coarse approximation of the posterior. The filter suffers from high variance, frequent particle depletion, and biased representations of the true posterior. In many cases, it effectively behaves like a single-particle filter after resampling.

Chapter 5 — Problem 1

Let the state at time $t - 1$ be

$$x_{t-1} = (x, \dot{x}),$$

where x is the position and $\dot{x}$ is the velocity. Let the control input be the commanded acceleration u , and assume that the actual acceleration is

$$a = u + \varepsilon,$$

where

$$\varepsilon \sim \mathcal{N}(0, \sigma^2).$$

Also assume that this acceleration remains constant over the time interval Δt .

Under constant acceleration, the state at time t is

$$x' = x + \dot{x}\Delta t + \frac{1}{2}a(\Delta t)^2,$$

$$\dot{x}' = \dot{x} + a\Delta t.$$

Substituting $a = u + \varepsilon$, we get

$$x' = x + \dot{x}\Delta t + \frac{1}{2}(u + \varepsilon)(\Delta t)^2,$$

$$\dot{x}' = \dot{x} + (u + \varepsilon)\Delta t.$$

To compute the posterior probability for a candidate pair $(x', \dot{x}')$, we infer the acceleration that would be required to produce that transition. From the position equation,

$$\hat{a} = \frac{2}{(\Delta t)^2} (x' - x - \dot{x}\Delta t).$$

From the velocity equation, the same acceleration must also satisfy

$$\hat{a} = \frac{\dot{x}' - \dot{x}}{\Delta t}.$$

Thus, a candidate pair $(x', \dot{x}')$ is only possible if these two expressions agree, i.e.

$$\frac{2}{(\Delta t)^2} (x' - x - \dot{x}\Delta t) = \frac{\dot{x}' - \dot{x}}{\Delta t}.$$

If this consistency condition is satisfied, then the posterior is given by the Gaussian probability of the required acceleration error:

$$p(x_t | u_t, x_{t-1}) = p(x', \dot{x}' | x, \dot{x}, u) = \mathcal{N}(\hat{a} - u; 0, \sigma^2).$$

Equivalently,

$$p(x', \dot{x}' | x, \dot{x}, u) = \begin{cases} \mathcal{N}\left(\frac{2}{(\Delta t)^2}(x' - x - \dot{x}\Delta t) - u; 0, \sigma^2\right), & \text{if } \frac{2}{(\Delta t)^2}(x' - x - \dot{x}\Delta t) = \frac{\dot{x}' - \dot{x}}{\Delta t}, \\ 0, & \text{otherwise.} \end{cases}$$

This shows that the posterior over $(x', \dot{x}')$ is degenerate: all probability mass lies on the set of state pairs that are consistent with a single constant acceleration over the interval.

Yes, x' and $\dot{x}'$ are correlated in the posterior. The reason is that both depend on the same random acceleration noise term ε . If the actual acceleration is larger than commanded, then both the final velocity and the final position increase; if it is smaller, both decrease. Therefore, the uncertainty in x' and $\dot{x}'$ is not independent.

Chapter 5 — Problem 2

Let the motion model be

$$x' = x + \dot{x}\Delta t + \frac{1}{2}a(\Delta t)^2, \quad \dot{x}' = \dot{x} + a\Delta t,$$

where the actual acceleration is

$$a = u + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

and is assumed constant over the interval Δt .

We are asked to compute the posterior distribution

$$p(\dot{x}' \mid x', x, \dot{x}, u).$$

From the position equation,

$$x' = x + \dot{x}\Delta t + \frac{1}{2}a(\Delta t)^2,$$

we can solve explicitly for the acceleration:

$$a = \frac{2}{(\Delta t)^2} (x' - x - \dot{x}\Delta t).$$

Thus, once x , $\dot{x}$, and x' are known, the value of a is uniquely determined. Substituting this into the velocity equation gives

$$\dot{x}' = \dot{x} + a\Delta t = \dot{x} + \Delta t \cdot \frac{2}{(\Delta t)^2} (x' - x - \dot{x}\Delta t).$$

Simplifying,

$$\dot{x}' = \frac{2(x' - x)}{\Delta t} - \dot{x}.$$

Therefore, the posterior over the final velocity is

$$p(\dot{x}' \mid x', x, \dot{x}, u) = \delta\left(\dot{x}' - \left(\frac{2(x' - x)}{\Delta t} - \dot{x}\right)\right).$$

Equivalently, we may write

$$p(\dot{x}' \mid x', x, \dot{x}, u) = p(\dot{x}' \mid x', x, \dot{x}),$$

because once x' , x , and $\dot{x}$ are given, the final velocity is completely determined and no longer depends on the commanded acceleration u .

The remarkable property of this posterior is that it is *degenerate*: it is not a Gaussian distribution with nonzero variance, but a Dirac delta concentrated at a single value. This happens because the model contains only one source of randomness, namely the acceleration noise ε , and the observed final position x' is sufficient to determine that noise exactly. Hence, after conditioning on x' , no uncertainty remains in $\dot{x}'$.

Chapter 5 — Problem 4

Let the bicycle state at time t be

$$x_t = (x, y, \theta),$$

where (x, y) denotes the position of the center of the front wheel, and θ denotes the orientation of the bicycle frame. Let the control input be

$$u_t = (v, \alpha),$$

where v is the forward speed of the front wheel center, and α is the steering angle of the front wheel, see Figure. 1. Assume that both controls remain constant during the time interval Δt , but are corrupted by Gaussian noise:

$$\begin{aligned} \hat{v} &= v + \varepsilon_v, & \varepsilon_v &\sim \mathcal{N}(0, \sigma_v^2), \\ \hat{\alpha} &= \alpha + \varepsilon_\alpha, & \varepsilon_\alpha &\sim \mathcal{N}(0, \sigma_\alpha^2), \end{aligned}$$

with ε_v and ε_α independent.

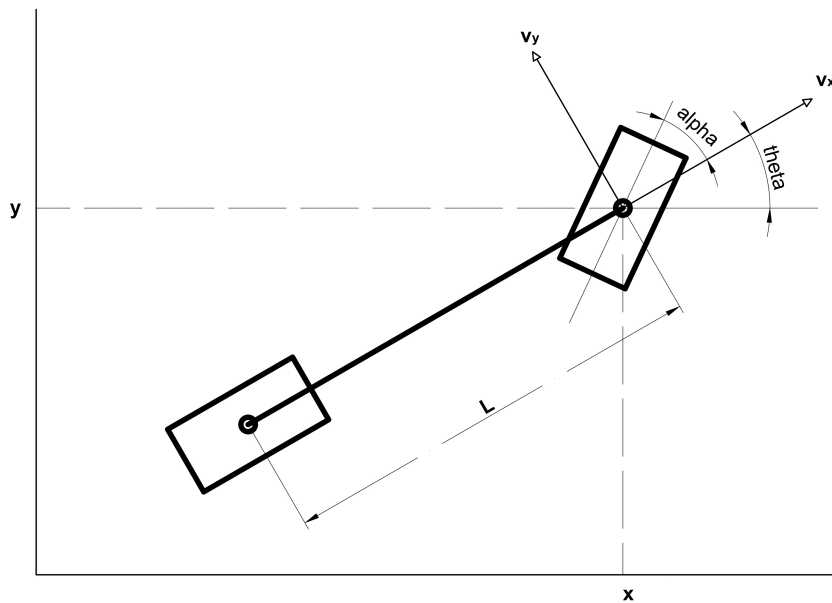


Figure 1: Demonstration of the bicycle model and parameters.

Under this convention, the front wheel center moves in the direction $\theta + \hat{\alpha}$, and the bicycle orientation evolves according to the geometric constraint imposed by the wheelbase L . The kinematic prediction model is therefore

$$\begin{aligned} x' &= x + \hat{v} \cos(\theta + \hat{\alpha}) \Delta t, \\ y' &= y + \hat{v} \sin(\theta + \hat{\alpha}) \Delta t, \\ \theta' &= \theta + \frac{\hat{v}}{L} \sin(\hat{\alpha}) \Delta t. \end{aligned}$$

Hence, the state transition function is

$$f(x, y, \theta, \hat{v}, \hat{\alpha}) = \begin{bmatrix} x + \hat{v} \cos(\theta + \hat{\alpha}) \Delta t \\ y + \hat{v} \sin(\theta + \hat{\alpha}) \Delta t \\ \theta + \frac{\hat{v}}{L} \sin(\hat{\alpha}) \Delta t \end{bmatrix}.$$

The exact prediction posterior is the distribution induced by the noisy controls:

$$p(x_t | x_{t-1}, u_t) = \iint \delta(x_t - f(x_{t-1}, \hat{v}, \hat{\alpha})) \mathcal{N}(\hat{v}; v, \sigma_v^2) \mathcal{N}(\hat{\alpha}; \alpha, \sigma_\alpha^2) d\hat{v} d\hat{\alpha}.$$

Because the mapping f is nonlinear in $\hat{v}$ and $\hat{\alpha}$, the exact posterior is generally not Gaussian. A first-order approximation is obtained by linearizing the motion model around the nominal controls (v, α) . Define the mean prediction

$$\mu = f(x, y, \theta, v, \alpha) = \begin{bmatrix} x + v \cos(\theta + \alpha) \Delta t \\ y + v \sin(\theta + \alpha) \Delta t \\ \theta + \frac{v}{L} \sin(\alpha) \Delta t \end{bmatrix}.$$

Let

$$w = \begin{bmatrix} \varepsilon_v \\ \varepsilon_\alpha \end{bmatrix}, \quad Q = \begin{bmatrix} \sigma_v^2 & 0 \\ 0 & \sigma_\alpha^2 \end{bmatrix}.$$

Using first-order linearization,

$$x_t \approx \mu + G w,$$

where G is the Jacobian of f with respect to the noisy controls:

$$G = \left[\frac{\partial f}{\partial \hat{v}} \quad \frac{\partial f}{\partial \hat{\alpha}} \right]_{(\hat{v}, \hat{\alpha}) = (v, \alpha)}.$$

Computing the derivatives gives

$$G = \begin{bmatrix} \cos(\theta + \alpha)\Delta t & -v \sin(\theta + \alpha)\Delta t \\ \sin(\theta + \alpha)\Delta t & v \cos(\theta + \alpha)\Delta t \\ \frac{\sin \alpha}{L}\Delta t & \frac{v \cos \alpha}{L}\Delta t \end{bmatrix}.$$

Therefore, the posterior may be approximated by the Gaussian

$$p(x_t | x_{t-1}, u_t) \approx \mathcal{N}(\mu, \Sigma),$$

with covariance

$$\Sigma = G Q G^T.$$

Thus, the exact prediction model is obtained by propagating the Gaussian noise in v and α through the nonlinear bicycle kinematics, and a practical approximation is to represent the resulting posterior by a Gaussian obtained through first-order linearization.

Chapter 6 — Problem 1

A ceiling is a good place for artificial markers because it is usually static, easy to view with an upward-looking camera, and less likely to be occluded by people or furniture.

Let

$$p_r = \begin{bmatrix} x_r \\ y_r \end{bmatrix}, \quad p_m = \begin{bmatrix} x_m \\ y_m \end{bmatrix}, \quad p_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}.$$

Define the planar rotation matrix by

$$R(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}.$$

We make the following coordinate choices.

- The world frame is a fixed frame parallel to the ceiling.
- The robot frame is attached to the robot, with heading θ_r .
- The camera is rigidly mounted on the robot, looks vertically upward, and its optical axis is perpendicular to the ceiling.
- The image coordinates are chosen so that the image x - and y -axes are parallel to the robot-frame x - and y -axes projected onto the ceiling.
- The marker lies on the ceiling plane at a constant distance h from the camera center.
- The camera center is taken to coincide with the robot reference point in the x - y plane.

With these choices, the marker position in the robot frame is

$$p_{m/r} = R(-\theta_r)(p_m - p_r).$$

By the given projective relation, a displacement in the ceiling plane is scaled by f/h in the image plane, hence

$$p_i = \frac{f}{h} p_{m/r} = \frac{f}{h} R(-\theta_r)(p_m - p_r).$$

Since the marker is symmetric under rotations of $\pi/2$, its orientation is observable only modulo $\pi/2$. Therefore,

$$\theta_i \equiv \theta_m - \theta_r \pmod{\pi/2}.$$

(a) **Marker global coordinates from image coordinates and robot pose**

From the projection equation,

$$p_m - p_r = R(\theta_r) \frac{h}{f} p_i,$$

so

$$p_m = p_r + R(\theta_r) \frac{h}{f} p_i$$

or, componentwise,

$$\begin{aligned} x_m &= x_r + \frac{h}{f} (x_i \cos \theta_r - y_i \sin \theta_r), \\ y_m &= y_r + \frac{h}{f} (x_i \sin \theta_r + y_i \cos \theta_r). \end{aligned}$$

For the orientation,

$$\theta_m \equiv \theta_r + \theta_i \pmod{\pi/2}.$$

(b) **Image coordinates from robot pose and marker coordinates**

Starting from the same relative-pose relation,

$$p_i = \frac{f}{h} R(-\theta_r)(p_m - p_r)$$

which gives

$$\begin{aligned} x_i &= \frac{f}{h} ((x_m - x_r) \cos \theta_r + (y_m - y_r) \sin \theta_r), \\ y_i &= \frac{f}{h} (-(x_m - x_r) \sin \theta_r + (y_m - y_r) \cos \theta_r). \end{aligned}$$

For the orientation,

$$\theta_i \equiv \theta_m - \theta_r \pmod{\pi/2}.$$

(c) **Robot pose from true marker coordinates and image coordinates**

From part (b),

$$\theta_i \equiv \theta_m - \theta_r \pmod{\pi/2},$$

so the robot orientation is not unique with this symmetric marker. The four possible orientations are

$$\theta_r^{(k)} = \theta_m - \theta_i + k \frac{\pi}{2}, \quad k \in \{0, 1, 2, 3\}.$$

For each candidate orientation,

$$p_r = p_m - R(\theta_r) \frac{h}{f} p_i,$$

therefore the four possible robot positions are

$$p_r^{(k)} = p_m - R(\theta_r^{(k)}) \frac{h}{f} p_i, \quad k \in \{0, 1, 2, 3\}.$$

Componentwise,

$$\begin{aligned} x_r^{(k)} &= x_m - \frac{h}{f} (x_i \cos \theta_r^{(k)} - y_i \sin \theta_r^{(k)}), \\ y_r^{(k)} &= y_m - \frac{h}{f} (x_i \sin \theta_r^{(k)} + y_i \cos \theta_r^{(k)}). \end{aligned}$$

Hence a single marker of the shown type does not determine a unique robot pose. It yields four pose hypotheses.

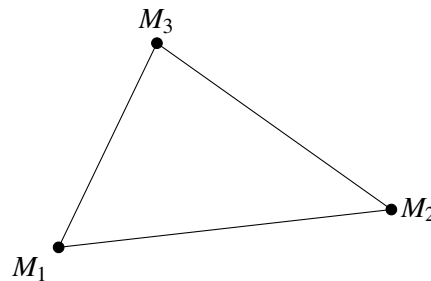
(d) **How many indistinguishable markers are needed for a unique pose?**

Answer: three markers are sufficient, provided they are arranged in a non-collinear and asymmetric configuration, for example a scalene triangle.

Two markers are not sufficient. Since the markers are indistinguishable, swapping the correspondence between the two observed image points and the two world markers produces another valid solution. Geometrically, this introduces a 180° ambiguity.

Three markers arranged as a scalene triangle are sufficient. The three pairwise distances uniquely identify the correspondence between image points and world markers, because a scalene triangle has no nontrivial symmetry. Once the correspondence is known, the planar rigid transform between the ceiling configuration and the image configuration is unique, so the robot pose is unique.

A sufficient configuration is shown below.



Three indistinguishable markers placed at the vertices of a scalene triangle.

Mathematically, if the three marker centers are $p_m^{(1)}, p_m^{(2)}, p_m^{(3)}$ and the three image points are $p_i^{(1)}, p_i^{(2)}, p_i^{(3)}$, then the unique pose (p_r, θ_r) satisfies

$$p_i^{(j)} = \frac{f}{h} R(-\theta_r) (p_m^{(j)} - p_r), \quad j = 1, 2, 3,$$

after the unique correspondence has been determined from the asymmetric triangle geometry.

Chapter 7 — Problem 1

Let the robot pose be

$$x_t = [x \quad y \quad \theta]^T,$$

and let the i -th observed landmark have known identity j_i and map coordinates

$$m_{j_i} = [m_{j_i,x} \quad m_{j_i,y}]^T.$$

Since the identity sensor is noise-free, we only keep range and bearing. From Chapter 6.6, the measurement model is

$$z_t^i = \begin{bmatrix} r_t^i \\ \phi_t^i \end{bmatrix} = h_i(x_t) + v_i, \quad v_i \sim \mathcal{N}(0, Q_i),$$

with

$$h_i(x_t) = \begin{bmatrix} \sqrt{(m_{j_i,x} - x)^2 + (m_{j_i,y} - y)^2} \\ \text{atan2}(m_{j_i,y} - y, m_{j_i,x} - x) - \theta \end{bmatrix}, \quad Q_i = \begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\phi^2 \end{bmatrix}.$$

Assuming conditional independence of the landmark measurements,

$$p(z_t^{1:k} | x_t, m) = \prod_{i=1}^k p(z_t^i | x_t, m).$$

- (a) **Why a single landmark is not well approximated by a Gaussian, whereas two or more often are.**

For a single landmark measurement (r, ϕ) of landmark m_j , the noise-free constraints are

$$r = \sqrt{(m_{j,x} - x)^2 + (m_{j,y} - y)^2}, \quad \phi = \text{atan2}(m_{j,y} - y, m_{j,x} - x) - \theta.$$

Equivalently,

$$x = m_{j,x} - r \cos(\theta + \phi), \quad y = m_{j,y} - r \sin(\theta + \phi).$$

Hence θ is free, and for every θ there is a corresponding (x, y) . Therefore the posterior is concentrated on a one-dimensional manifold in the three-dimensional pose space. In the x - y plane, this becomes a circle of radius r around the landmark. Such a ring-shaped or spiral-shaped density is strongly non-Gaussian.

When two or more landmarks are observed simultaneously, the posterior becomes

$$p(x_t | z_t^{1:k}, m) \propto \prod_{i=1}^k p(z_t^i | x_t, m).$$

Each measurement contributes another range-bearing constraint. For nondegenerate landmark geometry, the intersection of these constraints reduces the continuum of feasible poses to a small neighborhood around a single pose. Around that pose, the nonlinear measurement model can be linearized and the negative log-likelihood becomes approximately quadratic. Hence the posterior is often well approximated by a Gaussian.

- (b) **Procedure for obtaining a Gaussian pose estimate under a uniform prior.**

Because the initial prior is uniform,

$$p(x_t | z_t^{1:k}, m) \propto p(z_t^{1:k} | x_t, m).$$

A convenient way to construct an initial Gaussian is:

- (a) Choose one observed landmark, say landmark j_1 , as a proposal landmark.

(b) Sample M poses from the single-landmark posterior

$$q(x_t) = p(x_t | z_t^1, m) \propto p(z_t^1 | x_t, m),$$

using the inverse sensor model from Chapter 6.6. For sample index n :

$$\begin{aligned} \gamma^{[n]} &\sim \mathcal{U}(0, 2\pi), \\ \hat{r}^{[n]} &= r_t^1 + \varepsilon_r^{[n]}, \quad \varepsilon_r^{[n]} \sim \mathcal{N}(0, \sigma_r^2), \\ \hat{\phi}^{[n]} &= \phi_t^1 + \varepsilon_\phi^{[n]}, \quad \varepsilon_\phi^{[n]} \sim \mathcal{N}(0, \sigma_\phi^2), \\ x^{[n]} &= m_{j_1, x} + \hat{r}^{[n]} \cos \gamma^{[n]}, \\ y^{[n]} &= m_{j_1, y} + \hat{r}^{[n]} \sin \gamma^{[n]}, \\ \theta^{[n]} &= \gamma^{[n]} - \pi - \hat{\phi}^{[n]}. \end{aligned}$$

(c) Weight each sample using the remaining $k - 1$ landmark measurements:

$$\tilde{w}^{[n]} = \prod_{i=2}^k p(z_t^i | x_t^{[n]}, m),$$

where

$$p(z_t^i | x_t^{[n]}, m) \propto \exp\left(-\frac{1}{2} e_i(x_t^{[n]})^T Q_i^{-1} e_i(x_t^{[n]})\right),$$

and

$$e_i(x_t^{[n]}) = \begin{bmatrix} r_t^i - \hat{r}_i(x_t^{[n]}) \\ \text{wrap}(\phi_t^i - \hat{\phi}_i(x_t^{[n]})) \end{bmatrix}.$$

Normalize:

$$w^{[n]} = \frac{\tilde{w}^{[n]}}{\sum_{\ell=1}^M \tilde{w}^{[\ell]}}.$$

(d) Compute the weighted mean. For position,

$$\mu_x = \sum_{n=1}^M w^{[n]} x^{[n]}, \quad \mu_y = \sum_{n=1}^M w^{[n]} y^{[n]}.$$

For the angle, use the circular mean:

$$\mu_\theta = \text{atan2}\left(\sum_{n=1}^M w^{[n]} \sin \theta^{[n]}, \sum_{n=1}^M w^{[n]} \cos \theta^{[n]}\right).$$

Thus,

$$\mu = \begin{bmatrix} \mu_x \\ \mu_y \\ \mu_\theta \end{bmatrix}.$$

(e) Compute the weighted covariance

$$\Sigma = \sum_{n=1}^M w^{[n]} \delta^{[n]} (\delta^{[n]})^T,$$

with

$$\delta^{[n]} = \begin{bmatrix} x^{[n]} - \mu_x \\ y^{[n]} - \mu_y \\ \text{wrap}(\theta^{[n]} - \mu_\theta) \end{bmatrix}.$$

The resulting approximation is

$$p(x_t \mid z_t^{1:k}, m) \approx \mathcal{N}(x_t; \mu, \Sigma).$$

This Gaussian can then be used as the initial belief for the EKF. The method works well when the weighted samples cluster around a single dominant mode, which is the typical case when two or more nondegenerately placed landmarks are observed simultaneously.

Chapter 7 — Problem 3

The robot has a perfect compass, so its orientation is known exactly and does not need to be estimated. Therefore, the EKF state contains only position:

$$x_t = \begin{bmatrix} x_t \\ y_t \\ z_t \end{bmatrix}.$$

The control input is the commanded Cartesian velocity

$$u_t = \begin{bmatrix} \dot{x}_t \\ \dot{y}_t \\ \dot{z}_t \end{bmatrix},$$

and the sampling period is Δt .

(a) Derivation of the EKF localization algorithm

Since the robot moves independently in the three Cartesian directions, the deterministic motion model is

$$g(u_t, x_{t-1}) = x_{t-1} + \Delta t u_t.$$

With additive Gaussian motion noise,

$$x_t = g(u_t, x_{t-1}) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R_t),$$

where, by the problem statement, the noise is independent in the three directions:

$$R_t = \begin{bmatrix} \sigma_{x,t}^2 & 0 & 0 \\ 0 & \sigma_{y,t}^2 & 0 \\ 0 & 0 & \sigma_{z,t}^2 \end{bmatrix}.$$

Hence,

$$\boxed{x_t = x_{t-1} + \Delta t u_t + \varepsilon_t.}$$

The EKF uses a first-order Taylor expansion around the current mean μ_{t-1} :

$$g(u_t, x_{t-1}) \approx g(u_t, \mu_{t-1}) + G_t(x_{t-1} - \mu_{t-1}),$$

where

$$G_t = \left. \frac{\partial g}{\partial x} \right|_{\mu_{t-1}}.$$

Since $g(u_t, x_{t-1}) = x_{t-1} + \Delta t u_t$ is already linear,

$$\boxed{G_t = I_3.}$$

Therefore, the prediction equations are

$$\boxed{\bar{\mu}_t = \mu_{t-1} + \Delta t u_t}$$

and

$$\bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + R_t = \Sigma_{t-1} + R_t.$$

Let beacon j be located at the known point

$$m_j = \begin{bmatrix} m_{j,x} \\ m_{j,y} \\ m_{j,z} \end{bmatrix}.$$

Each beacon emits an acoustic signal at a known time τ_j^{emit} , and the robot measures the arrival time $\tau_{j,t}^{\text{arr}}$. If c is the speed of sound in water, then the measured range is

$$z_{j,t} = c(\tau_{j,t}^{\text{arr}} - \tau_j^{\text{emit}}).$$

Since the robot cannot measure the direction of arrival, each beacon provides only a scalar range. The nonlinear measurement model is therefore

$$z_{j,t} = h_j(x_t) + \delta_{j,t}, \quad \delta_{j,t} \sim \mathcal{N}(0, \sigma_{r,j}^2),$$

with

$$h_j(x_t) = \sqrt{(x_t - m_{j,x})^2 + (y_t - m_{j,y})^2 + (z_t - m_{j,z})^2}.$$

For k simultaneous beacon measurements, define

$$z_t = \begin{bmatrix} z_{1,t} \\ \vdots \\ z_{k,t} \end{bmatrix}, \quad h(x_t) = \begin{bmatrix} h_1(x_t) \\ \vdots \\ h_k(x_t) \end{bmatrix}, \quad Q_t = \text{diag}(\sigma_{r,1}^2, \dots, \sigma_{r,k}^2).$$

Thus,

$$z_t = h(x_t) + \delta_t, \quad \delta_t \sim \mathcal{N}(0, Q_t).$$

The EKF linearizes $h(x_t)$ around the predicted mean $\bar{\mu}_t$:

$$h(x_t) \approx h(\bar{\mu}_t) + H_t(x_t - \bar{\mu}_t),$$

where

$$H_t = \frac{\partial h}{\partial x} \Big|_{\bar{\mu}_t}.$$

For beacon j , let

$$d_j(\bar{\mu}_t) = \sqrt{(\bar{\mu}_{t,x} - m_{j,x})^2 + (\bar{\mu}_{t,y} - m_{j,y})^2 + (\bar{\mu}_{t,z} - m_{j,z})^2}.$$

Then the j -th row of H_t is

$$\frac{\partial h_j}{\partial x} \Big|_{\bar{\mu}_t} = \begin{bmatrix} \frac{\bar{\mu}_{t,x} - m_{j,x}}{d_j(\bar{\mu}_t)} & \frac{\bar{\mu}_{t,y} - m_{j,y}}{d_j(\bar{\mu}_t)} & \frac{\bar{\mu}_{t,z} - m_{j,z}}{d_j(\bar{\mu}_t)} \end{bmatrix}.$$

Therefore,

$$H_t = \begin{bmatrix} \frac{\bar{\mu}_{t,x} - m_{1,x}}{d_1(\bar{\mu}_t)} & \frac{\bar{\mu}_{t,y} - m_{1,y}}{d_1(\bar{\mu}_t)} & \frac{\bar{\mu}_{t,z} - m_{1,z}}{d_1(\bar{\mu}_t)} \\ \vdots & \vdots & \vdots \\ \frac{\bar{\mu}_{t,x} - m_{k,x}}{d_k(\bar{\mu}_t)} & \frac{\bar{\mu}_{t,y} - m_{k,y}}{d_k(\bar{\mu}_t)} & \frac{\bar{\mu}_{t,z} - m_{k,z}}{d_k(\bar{\mu}_t)} \end{bmatrix}.$$

Using the predicted mean,

$$\hat{z}_t = h(\bar{\mu}_t).$$

The innovation covariance is

$$S_t = H_t \bar{\Sigma}_t H_t^T + Q_t,$$

and the Kalman gain is

$$K_t = \bar{\Sigma}_t H_t^T S_t^{-1}.$$

Then the corrected state estimate and covariance are

$$\mu_t = \bar{\mu}_t + K_t(z_t - \hat{z}_t)$$

and

$$\Sigma_t = (I - K_t H_t) \bar{\Sigma}_t.$$

Assuming known correspondence, one EKF iteration is:

Input: $\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, \{m_j\}, R_t, Q_t$

Prediction:

$$\bar{\mu}_t = \mu_{t-1} + \Delta t u_t$$

$$\bar{\Sigma}_t = \Sigma_{t-1} + R_t$$

Measurement prediction:

$$\hat{z}_t = h(\bar{\mu}_t)$$

$$H_t = \left. \frac{\partial h}{\partial x} \right|_{\bar{\mu}_t}$$

Correction:

$$S_t = H_t \bar{\Sigma}_t H_t^T + Q_t$$

$$K_t = \bar{\Sigma}_t H_t^T S_t^{-1}$$

$$\mu_t = \bar{\mu}_t + K_t(z_t - \hat{z}_t)$$

$$\Sigma_t = (I - K_t H_t) \bar{\Sigma}_t$$

Output: μ_t, Σ_t

(b) Simulation and investigation of accuracy and failure modes

The EKF derived in part (a) was implemented in simulation for the three localization settings requested in the problem: position tracking, global localization, and the kidnapped robot problem. The environment contained four known acoustic beacons, and the robot received range-only measurements from them. The beacon identities were assumed known.

The filter was initialized near the true position with a small covariance. In this case, the EKF tracks the robot accurately, as shown in Figure 2. The true and estimated trajectories remain close throughout the run.

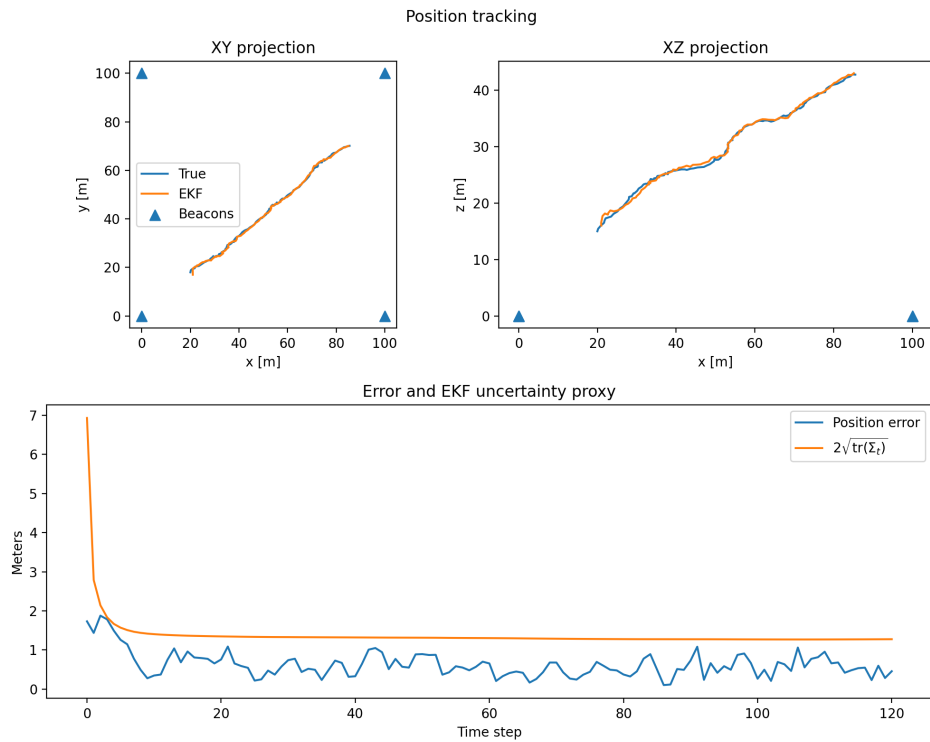


Figure 2: Position tracking case. The EKF is initialized close to the true state and produces accurate local tracking.

The filter was initialized far from the true state. Since the measurements are ranges only, and the four beacons are coplanar, the problem admits ambiguous mirror-like solutions in the vertical direction. A single Gaussian belief cannot represent multiple hypotheses. As a result, the EKF converges to the wrong mode in this test, as shown in Figure 3.

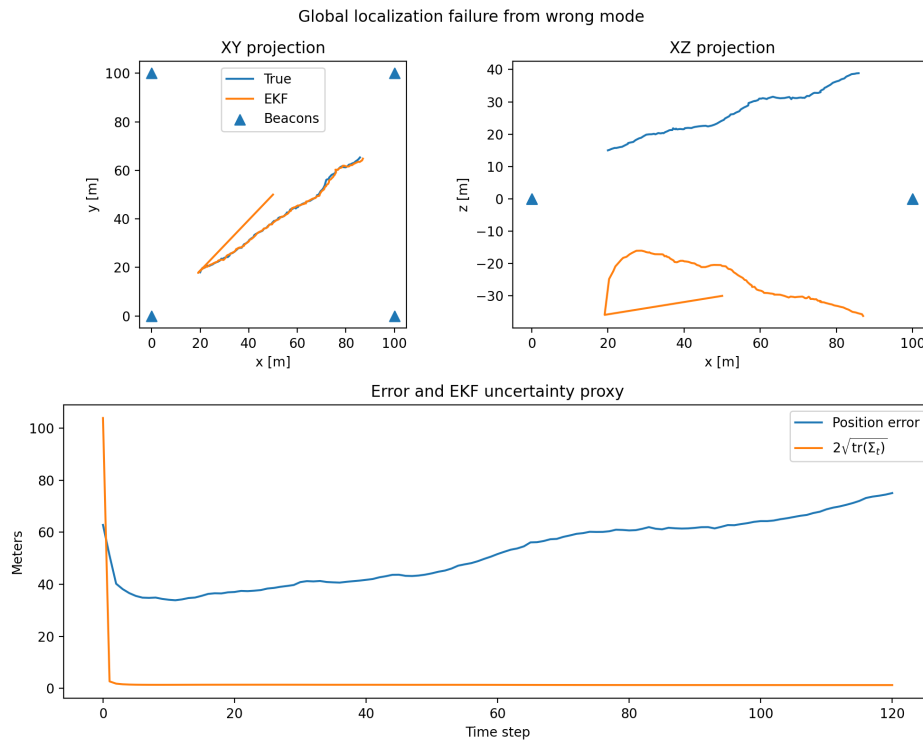


Figure 3: Global localization case. The EKF fails because the true posterior is not well represented by a single Gaussian under range-only ambiguous geometry.

The filter first tracks the robot correctly, then the true robot position is changed abruptly while the filter remains unaware of that jump. The error increases sharply after kidnapping, then decreases gradually as new measurements are incorporated. This behavior is shown in Figure 4.

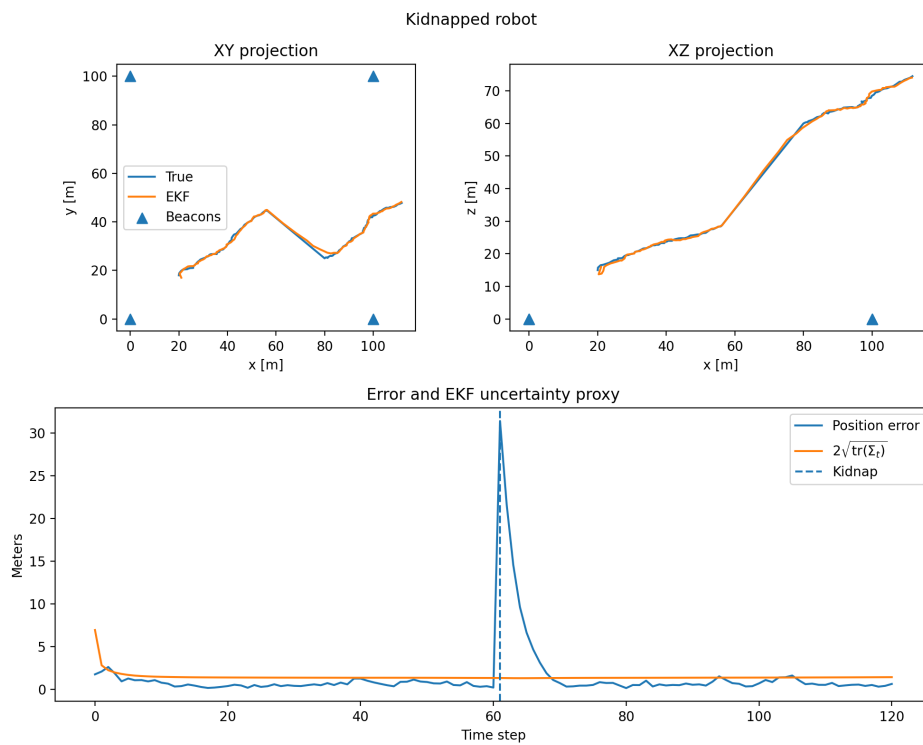


Figure 4: Kidnapped robot case. The EKF becomes temporarily inconsistent after the abrupt relocation, then gradually re-converges.

To compare the three scenarios directly, the simulation also computed Monte Carlo RMSE statistics over repeated runs. Figure 5 summarizes the error levels. As expected, position tracking is the easiest case, global localization is the most difficult, and the kidnapped robot problem lies in between because the filter must recover after a sudden state jump.

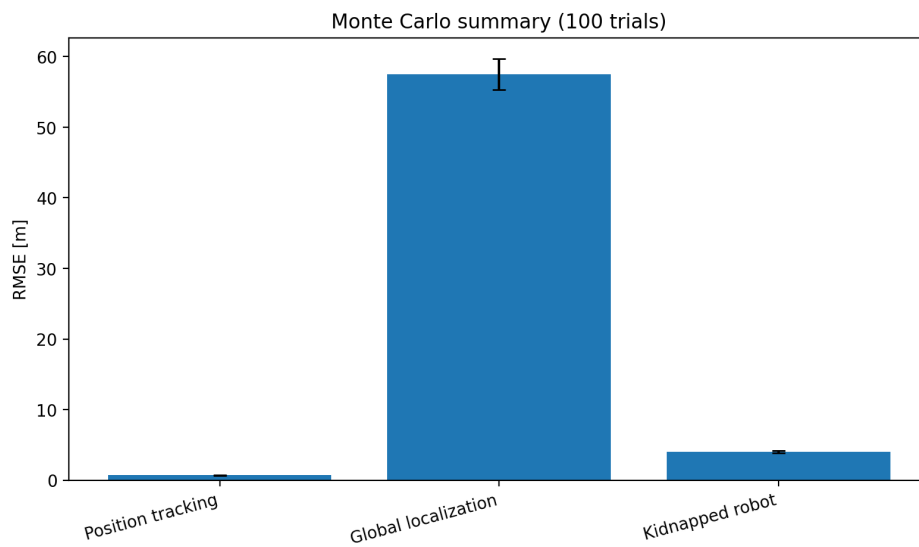


Figure 5: Monte Carlo RMSE summary for the three scenarios.